LESSON 1-2

Transformations of Functions

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Program Resources

Lesson Overview

FOCUS

Objective

Students will be able to:

- ✔ Graph a transformed function by identifying the effect on the graph of replacing $f(x)$ by $f(x) + k$, $kf(x)$, $f(kx)$, and $f(x + k)$ for specific values of k.
- ✔ Write an equation of a transformed function.
- ✔ Relate the domain of a function to its graph and the real-world situation it describes.

Essential Understanding

A function of the form $f(x) = a \cdot f[b(x - h)] + k$ is transformed by changing the values of a, b, h, or k. Changing the value of h or k results in a horizontal or vertical translation, changing the sign of a or b results in a reflection across one of the axes, and changing the value of a or b results in a horizontal or vertical stretch or compression.

COHERENCE

Previously in earlier courses, students:

- Created, graphed, and identified the key features of the graphs of linear, quadratic, and absolute value functions.

In this lesson, students:

- Write equations for linear and quadratic functions by understanding how changing the values of a, b, h, and k affect the key features of the graph of a function.

Later in this topic, students will:

- Create, graph, and identify the key features of piecewise-defined functions such as absolute value functions and step functions.

RIGOR

This lesson emphasizes a blend of *conceptual understanding* and *application*.

- Students apply their understanding of transformations either to graph a translation of a linear, quadratic, or absolute value function or to write an equation given a graph.

Glossary is available at SavvasRealize.com.

Vocabulary Builder

NEW VOCABULARY

English | *Spanish*

- **compression** | *compresión*
- **reflection** | *reflexión*
- **stretch** | *tramo*
- **transformation** | *transformación*
- **translation** | *translación*

VOCABULARY ACTIVITY

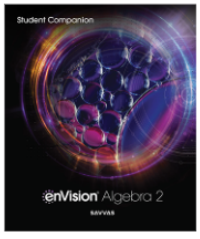
Using a piece of clay or similar material, provide a visual example of the terms *compression* and *stretch*. Tell students that a transformation of a function in the form $f(x) = a \cdot f[b(x - h)] + k$ involves a change to at least one of the values a, b, h, or k. Tell them that some transformations affect the vertical distance between the points of the graph of a function. When the points are closer together, it is called a *compression,* and when the points are further apart, it is called a *stretch.* Have students complete the following sentences.

A transformation that decreases the distance between the points of the graph is called a _____. **[compression]**

A transformation that increases the distance between the points of the graph is called a _____. **[stretch]**

Student Companion

Students can do their work for the lesson in their *Student Companion* or on Savvas Realize.



Mathematics Overview

Students interpret, graph, and write equations of transformations of functions. Functions in the form $y = f(x - h)$ represent horizontal translations, functions in the form $y = f(x) + k$ represent vertical translation, functions in the form $y = -f(x)$ represent reflections across the x-axis, functions in the form $y = f(-x)$ represent reflections across the y-axis, functions in the form $y = f(bx)$ represent horizontal stretches or compressions, and functions in the form $y = af(x)$ represent vertical stretches or compressions.

Applying Math Practices

Use Appropriate Tools Strategically

Students use available tools, such as graphing calculators, to graph the original and transformed equations to check that the transformations they identified are correct.

Look For and Make Use of Structure

Students look for overall structure and patterns in mathematics when identifying why the transformation $g(x) = -f(x)$ affects the y-coordinate of each point and why the transformation $g(x) = f(-x)$ affects the x-coordinate of each point.

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Activity

EXPLORE & REASON

INSTRUCTIONAL FOCUS Students explore the graph of an absolute value function and how different values for c impact the graph of the function.

STUDENT COMPANION Students can complete the *Explore & Reason* activity in their *Student Companion*.

Before WHOLE CLASS

Implement Tasks that Promote Reasoning and Problem Solving ETP

Q: What does the graph of the function tell you about the range of $f(x) = |x|$?
[The range is greater than or equal to zero.]

Q: What can you predict about the range of the graphs of the function $g(x) = |x + c|$ for different values of c?
[Regardless of the value of c, the range will still be greater than or equal to zero.]

During SMALL GROUP

Support Productive Struggle in Learning Mathematics ETP

Q: How can you find points to graph for each function?
[Make a table of values using values of x between −5 and 5 and calculate the corresponding y-values.]

Q: What do you notice about the vertex of each function?
[The vertices all lie on the x-axis.]

For Early Finishers

Q: How does the graph change if the coefficient of $f(x + c)$ is negative? Choose three of your functions and graph $-f(x + c)$.
[The graphs of $-f(x + c)$ are reflections across the x-axis of the graphs of $f(x + c)$.]

After WHOLE CLASS

Facilitate Meaningful Mathematical Discourse ETP

Discuss how transformations applied to functions can affect the range, the domain, or both.

Q: If the function $f(x)$ has a horizontal shift, what part of the function has been changed?
[In the function $f(x)$, x stands for the domain; therefore, operations will be applied to x within the parentheses.]

Q: If the function $f(x)$ has a vertical shift, what part of the function has been changed?
[In the function $f(x)$, $f(x)$ stands for the range; therefore, operations will be applied to $f(x)$.]

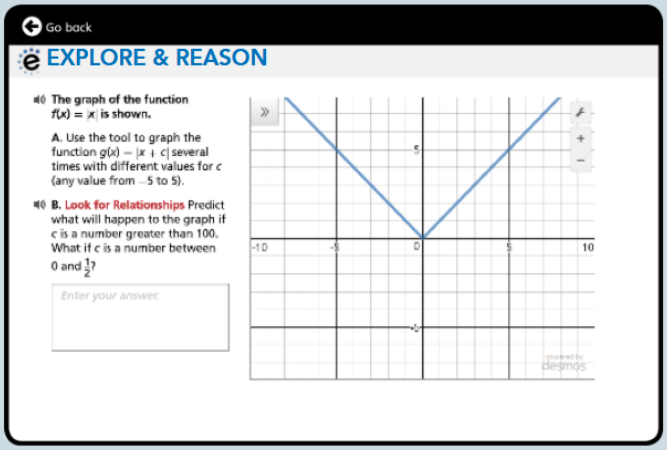
Introduce students to the language of functions written in the form $f(x) = a \cdot f[b(x - h)] + k$ and describe how that notation is used to define certain movements of the graph of a function.

Use with EXPLORE & REASON

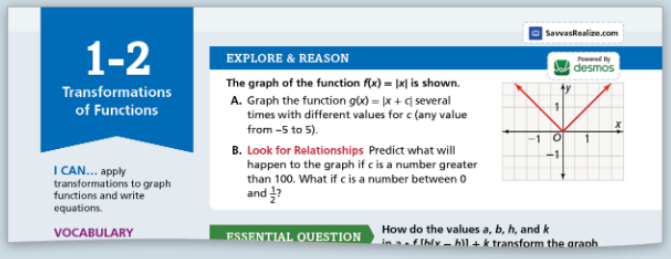
HABITS OF MIND
Reason What happens to the graph if c is a negative number?

[The graph of the function is shifted to the right when c is a negative number.]

Explore & Reason is Powered by Desmos.
Activities are available at SavvasRealize.com.



STUDENT EDITION



SAMPLE STUDENT WORK

A. graphs with several absolute value functions identical to $f(x)=|x|$ but shifted several units to the left or right

B. For $c > 100$, you will see a very large horizontal shift of the graph to the left. For $0 < c < \frac{1}{2}$ you will see a very small horizontal shift of the graph to the left.

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Activity

INTRODUCE THE ESSENTIAL QUESTION

Establish Mathematics Goals to Focus Learning ETP

Introduce students to the different types of transformations of functions. Remind them that transformations are defined by vertical and horizontal translations, reflections, and stretches and compressions on the coordinate plane. Students can apply these transformations to graphing functions and writing equations.

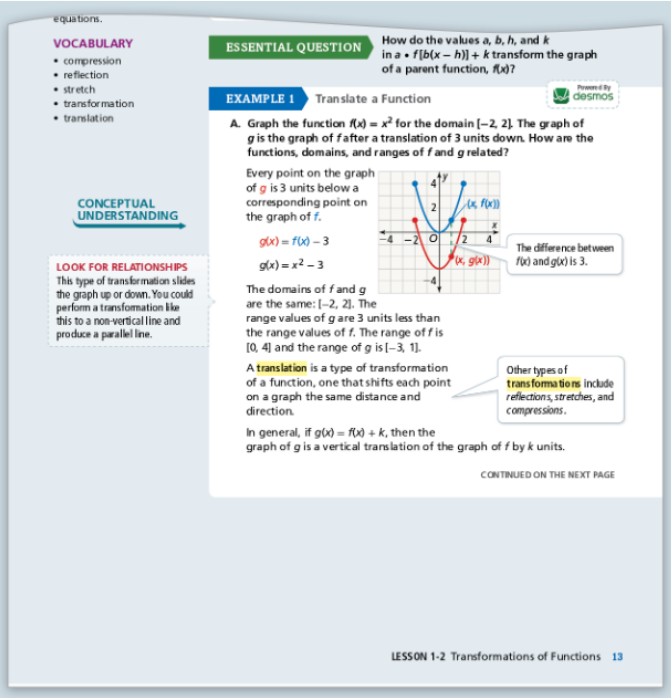
EXAMPLE 1 Translate a Function

Use and Connect Mathematical Representations ETP

Q: What do you notice about the two new graphs compared to their parent functions?
[One graph is the parent function translated down; the other graph is the parent function translated to the right.]

Q: Will the transformation of either graph cause the parabola to change shape?
[No; a translation is a slide to a new position. The shape will not change.]

Example 1 is Powered by Desmos.
Examples are available at SavvasRealize.com.


equations.

VOCABULARY
- compression
- reflection
- stretch
- transformation
- translation

ESSENTIAL QUESTION How do the values a, b, h, and k in $a \cdot f[b(x - h)] + k$ transform the graph of a parent function, $f(x)$?

EXAMPLE 1 Translate a Function

A. Graph the function $f(x) = x^2$ for the domain $[-2, 2]$. The graph of g is the graph of f after a translation of 3 units down. How are the functions, domains, and ranges of f and g related?

CONCEPTUAL UNDERSTANDING

LOOK FOR RELATIONSHIPS
This type of transformation slides the graph up or down. You could perform a transformation like this to a non-vertical line and produce a parallel line.

Every point on the graph of g is 3 units below a corresponding point on the graph of f.

$g(x) = f(x) - 3$

$g(x) = x^2 - 3$

The difference between $f(x)$ and $g(x)$ is 3.

The domains of f and g are the same: $[-2, 2]$. The range values of g are 3 units less than the range values of f. The range of f is $[0, 4]$ and the range of g is $[-3, 1]$.

A translation is a type of transformation of a function, one that shifts each point on a graph the same distance and direction.

Other types of transformations include *reflections*, *stretches*, and *compressions*.

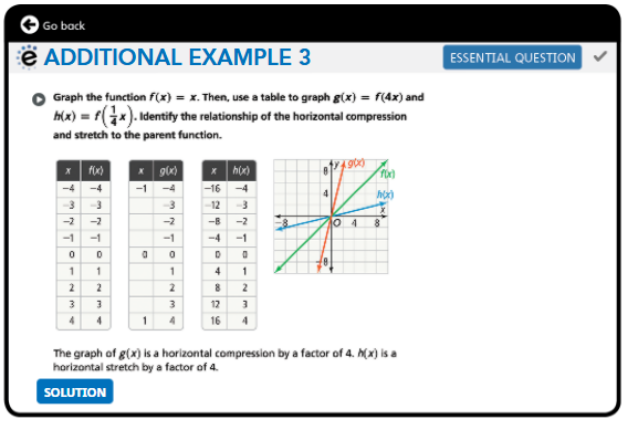
In general, if $g(x) = f(x) + k$, then the graph of g is a vertical translation of the graph of f by k units.

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LESSON 1-2 Transformations of Functions 13

Additional Examples

Additional Examples are available at SavvasRealize.com.


Go back

ADDITIONAL EXAMPLE 3 ESSENTIAL QUESTION

Graph the function $f(x) = x$. Then, use a table to graph $g(x) = f(4x)$ and $h(x) = f\left(\frac{1}{4}x\right)$. Identify the relationship of the horizontal compression and stretch to the parent function.

x	$f(x)$
−4	−4
−3	−3
−2	−2
−1	−1
0	0
1	1
2	2
3	3
4	4

x	$g(x)$
−1	−4
	−3
	−2
	−1
0	0
	1
	2
	3
1	4

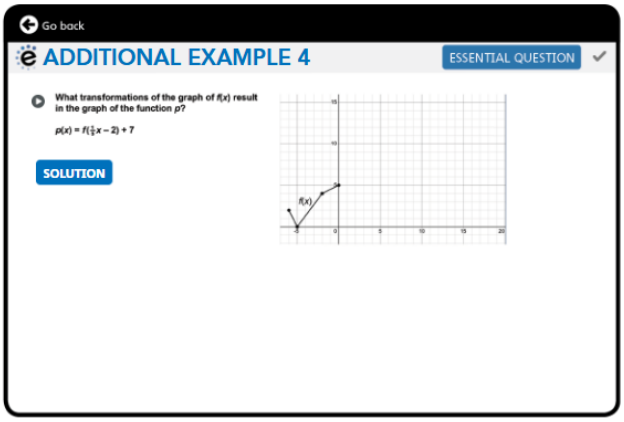
x	$h(x)$
−16	−4
−12	−3
−8	−2
−4	−1
0	0
4	1
8	2
12	3
16	4

The graph of $g(x)$ is a horizontal compression by a factor of 4. $h(x)$ is a horizontal stretch by a factor of 4.

SOLUTION

Example 3 Help students understand stretches and compressions of linear functions.

Q: What do you notice about the output values of $g(x)$ and $h(x)$ compared with the output values of $f(x)$?
[For the same input value, the output value of g is $\frac{1}{3}$ the output value of f and the output value of h is 3 times the output value of f.]


Go back

ADDITIONAL EXAMPLE 4 ESSENTIAL QUESTION

What transformations of the graph of $f(x)$ result in the graph of the function p?

$p(x) = f(\frac{1}{3}x - 2) + 7$

SOLUTION

Example 4 Help students understand functions that have more than two transformations. This example shows them how to organize an approach to graphing more complex functions.

Q: How can you determine the horizontal transformations of the graph of f?
[In the function, the coefficient of $\frac{1}{3}$ indicates a horizontal stretch. Factor $\frac{1}{3}$ to find the number subtracted from x, which indicates the horizontal translation.]

Try It! Answers

1. a. From f to g, the y-intercept is translated down 3 units. f had one x-intercept at the origin, and g has two x-intercepts on either side of the origin.
   b. From f to g, the x-intercept is translated right 3 units. f had one y-intercept at the origin, but g does not have a y-intercept.

Elicit and Use Evidence of Student Thinking ETP

Q: Will a horizontal translation always result in the translated function not having a y-intercept?
[No; if the domain of f were not restricted, then any horizontal translation of f would still have a y-intercept.]

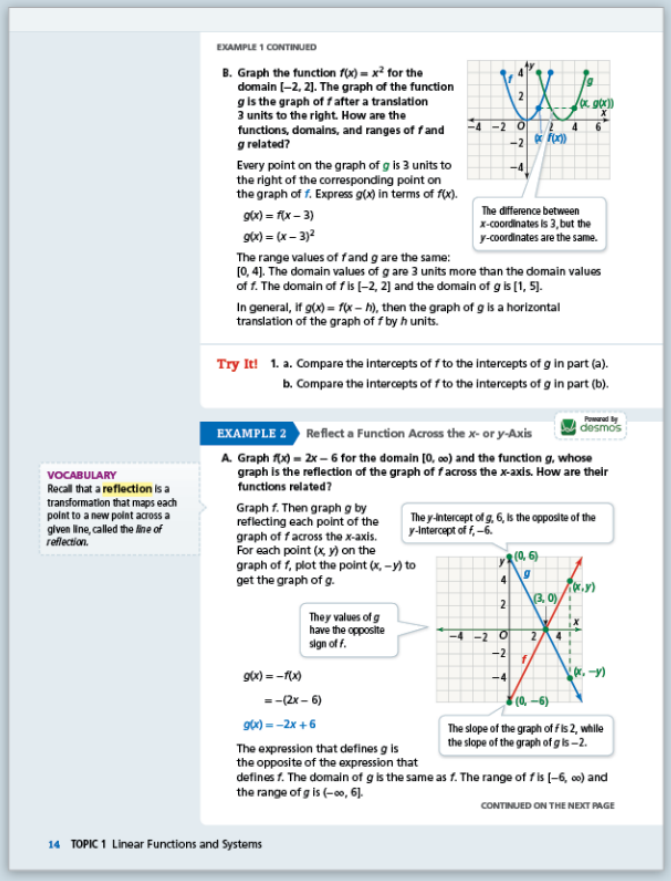
EXAMPLE 2 Reflect a Function Across the x- or y-Axis

Pose Purposeful Questions ETP

PART A

Q: How can you describe what the reflection across the x-axis does to the coordinates and the equation?
[The y-values of the original function are multiplied by -1.]

Example 2 is Powered by Desmos.
Examples are available at SavvasRealize.com.



EXAMPLE 1 CONTINUED

B. **Graph the function $f(x) = x^2$ for the domain $[-2, 2]$. The graph of the function g is the graph of f after a translation 3 units to the right. How are the functions, domains, and ranges of f and g related?**

Every point on the graph of g is 3 units to the right of the corresponding point on the graph of f. Express $g(x)$ in terms of $f(x)$.

$g(x) = f(x - 3)$

$g(x) = (x - 3)^2$

The difference between x-coordinates is 3, but the y-coordinates are the same.

The range values of f and g are the same: $[0, 4]$. The domain values of g are 3 units more than the domain values of f. The domain of f is $[-2, 2]$ and the domain of g is $[1, 5]$.

In general, if $g(x) = f(x - h)$, then the graph of g is a horizontal translation of the graph of f by h units.

Try It! 1. a. Compare the intercepts of f to the intercepts of g in part (a).
b. Compare the intercepts of f to the intercepts of g in part (b).

EXAMPLE 2 Reflect a Function Across the x- or y-Axis

VOCABULARY
Recall that a **reflection** is a transformation that maps each point to a new point across a given line, called the *line of reflection*.

A. **Graph $f(x) = 2x - 6$ for the domain $[0, \infty)$ and the function g, whose graph is the reflection of the graph of f across the x-axis. How are their functions related?**

Graph f. Then graph g by reflecting each point of the graph of f across the x-axis. For each point (x, y) on the graph of f, plot the point $(x, -y)$ to get the graph of g.

The y-intercept of g, 6, is the opposite of the y-intercept of f, -6.

The y values of g have the opposite sign of f.

$g(x) = -f(x)$

$= -(2x - 6)$

$g(x) = -2x + 6$

The slope of the graph of f is 2, while the slope of the graph of g is -2.

The expression that defines g is the opposite of the expression that defines f. The domain of g is the same as f. The range of f is $[-6, \infty)$ and the range of g is $(-\infty, 6]$.

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ELL English Language Learners (*Use with* EXAMPLE 2)

SPEAKING BEGINNING Ask students to tell what a reflection is. Then give each student a mirror and have them draw the reflections of different lines and shapes on a coordinate grid—some across the x-axis and some across the y-axis. Have them describe their reflections.

Q: How do the points of a graph change when reflected across the x-axis? The y-axis?
[The y-coordinates change to the opposite sign when reflected across the x-axis. The x-coordinates change to the opposite sign when reflected across the y-axis.]

Q: How are reflections related to symmetry?
[Symmetrical shapes can be divided by a line down the middle, with one side a reflection of the other side.]

LISTENING DEVELOPING Read the second paragraph of Part A aloud as students listen. Then play a game where students listen to directions about sending game pieces to different positions on a coordinate grid. For example, "Place the game piece on the point (4, 2). Now, send the piece to the opposite y-value."

Q: When you move from a positive y-value to a negative y-value, in which direction do you move?
[down]

Q: When you move from a negative x-value to a positive x-value, in which direction do you move?
[right]

WRITING EXPANDING Sometimes words are spelled the same but are pronounced differently and have different meanings. Point out the word *produce* in the example. Have students write two sentences using the word in two different ways.

Q: What topic could you write about if you used the word *produce* pronounced with the stress on the first syllable?
[fruits and vegetables; grocery store]

Q: What is a synonym for *produce* as it is used in the example?
[*make, create, form*]

PART B

Q: How can you describe what the reflection across the y-axis does to the coordinates and the equation?
[The x-values of the original function are multiplied by -1.]

Q: What is the difference between $g(x) = -f(x)$ and $g(x) = f(-x)$?
[When $f(x)$ has a coefficient of -1, the y-values change sign, causing a reflection across the x-axis. When x has a coefficient of -1, the x-values change sign, causing a reflection across the y-axis.]

Try It! Answers

2. a. $y = -x^2 + 2$
   b. $y = x^2 - 2$

Common Error

Try It! 2b Some students may incorrectly interpret a reflection across the y-axis as the result of changing the coefficient of x to -1. Show students that the x in the function is not changed to a negative; the actual values that are substituted into the function will be opposites of those from the original function but will produce the same range.

Use with EXAMPLES 1 & 2

HABITS OF MIND
Look for Relationships How are the intercepts of a graph affected by reflection across the x-axis? Explain.

[The x-intercept stays the same because the points on the axis you are reflecting about do not change. The y-intercept of the reflected graph is the opposite of the y-intercept of the original graph because the two points are equidistant from the x-axis.]

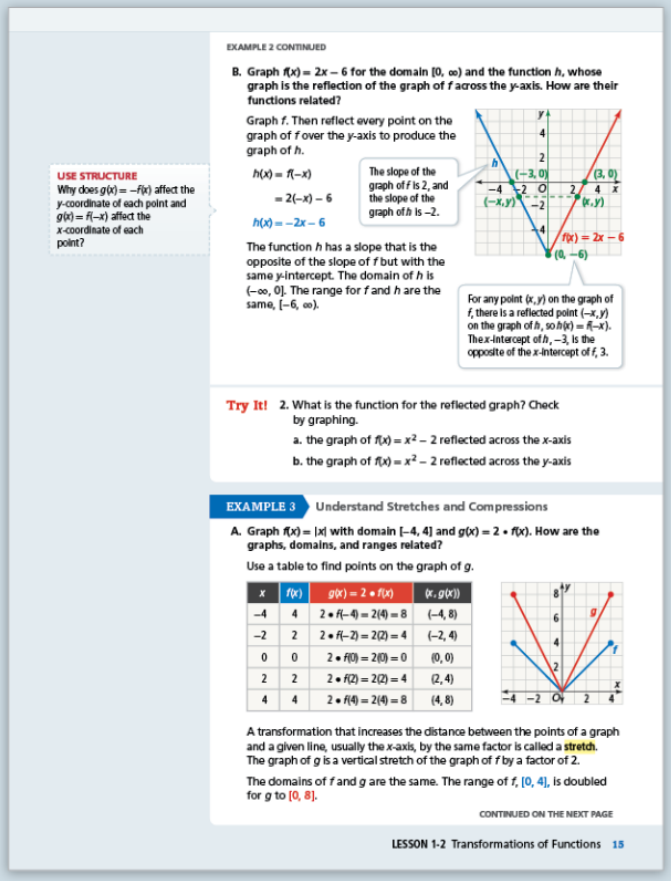
EXAMPLE 3 Understand Stretches and Compressions

Build Procedural Fluency From Conceptual Understanding ETP

PART A

Q: What is the relationship between the coefficient of $f(x) = a|x|$ and its effects on the graph?
[The coefficient of $f(x)$ is the factor by which the graph is either stretched or compressed vertically with regard to each point's distance from the x-axis.]

Examples are available at SavvasRealize.com.



EXAMPLE 2 CONTINUED

B. Graph $f(x) = 2x - 6$ for the domain $[0, \infty)$ and the function h, whose graph is the reflection of the graph of f across the y-axis. How are their functions related?

Graph f. Then reflect every point on the graph of f over the y-axis to produce the graph of h.

USE STRUCTURE
Why does $g(x) = -f(x)$ affect the y-coordinate of each point and $g(x) = f(-x)$ affect the x-coordinate of each point?

$h(x) = f(-x)$
$= 2(-x) - 6$
$h(x) = -2x - 6$

The slope of the graph of f is 2, and the slope of the graph of h is -2.

The function h has a slope that is the opposite of the slope of f but with the same y-intercept. The domain of h is $(-\infty, 0]$. The range for f and h are the same, $[-6, \infty)$.

For any point (x, y) on the graph of f, there is a reflected point $(-x, y)$ on the graph of h, so $h(x) = f(-x)$. The x-intercept of h, -3, is the opposite of the x-intercept of f, 3.

Try It! 2. What is the function for the reflected graph? Check by graphing.
a. the graph of $f(x) = x^2 - 2$ reflected across the x-axis
b. the graph of $f(x) = x^2 - 2$ reflected across the y-axis

EXAMPLE 3 Understand Stretches and Compressions

A. Graph $f(x) = |x|$ with domain $[-4, 4]$ and $g(x) = 2 \cdot f(x)$. How are the graphs, domains, and ranges related?

Use a table to find points on the graph of g.

x	$f(x)$	$g(x) = 2 \cdot f(x)$	$(x, g(x))$
-4	4	$2 \cdot f(-4) = 2(4) = 8$	$(-4, 8)$
-2	2	$2 \cdot f(-2) = 2(2) = 4$	$(-2, 4)$
0	0	$2 \cdot f(0) = 2(0) = 0$	$(0, 0)$
2	2	$2 \cdot f(2) = 2(2) = 4$	$(2, 4)$
4	4	$2 \cdot f(4) = 2(4) = 8$	$(4, 8)$

A transformation that increases the distance between the points of a graph and a given line, usually the x-axis, by the same factor is called a **stretch**. The graph of g is a vertical stretch of the graph of f by a factor of 2.

The domains of f and g are the same. The range of f, $[0, 4]$, is doubled for g to $[0, 8]$.

CONTINUED ON THE NEXT PAGE

LESSON 1-2 Transformations of Functions 15

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Activity

PART B

Q: What is the relationship between the coefficient in the function $f(x) = |bx|$ and the effects on the graph of the function?

[When $b > 1$, the graph is compressed horizontally by a factor $\frac{1}{b}$. When $b < 1$, the graph is stretched by $\frac{1}{b}$.]

Try It! Answer

3. If (x, y) is on the graph of j, then $\left(\frac{1}{2}x, y\right)$ is on the graph of f. So, every point on f corresponds to a point on j whose y-coordinate is the same and whose x-coordinate is doubled. So, the graph of j is a horizontal stretch of f. If $[a, b]$ is the domain of j, then $\left[\frac{1}{2}a, \frac{1}{2}b\right]$ is the domain of f.

Examples are available at SavvasRealize.com.

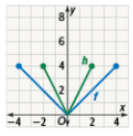
EXAMPLE 3 CONTINUED

B. Graph $f(x) = |x|$ with domain $[-4, 4]$ and $h(x) = f(2x)$. How are the graphs, domains, and ranges related?

Use a table to find points on the graph of h.

COMMON ERROR
Be careful not to assume that the domain of a transformed function is the same as the domain of the original function. Notice that $h(-4)$ would equal $f(-8)$, which is outside the domain of f, $[-4, 4]$.

x	$f(x)$	$h = f(2x)$	$(x, h(x))$
-2	2	$f(2(-2)) = f(-4) = 4$	$(-2, 4)$
-1	1	$f(2(-1)) = f(-2) = 2$	$(-1, 2)$
0	0	$f(2(0)) = f(0) = 0$	$(0, 0)$
1	1	$f(2(1)) = f(2) = 2$	$(1, 2)$
2	2	$f(2(2)) = f(4) = 4$	$(2, 4)$

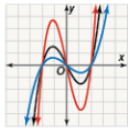
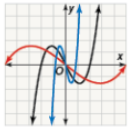


A transformation that decreases the distance between the points of a graph and a given line, usually the y-axis, by the same factor is called a **compression**. The graph of h is a horizontal compression of the graph of f by a factor of 2.

For each corresponding output, the value of the input for h is half the value of the input for f. The two functions have the same range. The values in the domain of h, $[-2, 2]$, are half as large as the values in the domain of f, $[-4, 4]$.

Try It! 3. The domain of $j(x)$ is $[a, b]$. Show that $j(x) = f\left(\frac{1}{2}x\right)$ is a horizontal stretch of the graph of f. What is the domain of f?

CONCEPT Stretches and Compressions

Vertical	Horizontal
$g(x) = k \cdot f(x)$	$h(x) = f(kx)$
	
• stretch when $k > 1$ • compression when $0 < k < 1$	• stretch when $0 < k < 1$ • compression when $k > 1$

16 TOPIC 1 Linear Functions and Systems

Support Struggling Students

USE WITH EXAMPLE 3 For extra support, have students practice creating a table to find points of the function when the domain of f is $[-5, 5]$.

- Use the function $h(x) = \frac{1}{2}f(x)$ to build a table and describe the range and domain of the function.

Q: Describe the domain and range of h.

[The domain of h is the same as that of f, and the range is $\frac{1}{2}$ the range of f.]

Q: How will the transformation affect the points on the graph of, h when compared to the graph of f?

[Each corresponding point on the graph of h is half the distance from the x-axis as the points on the graph of f.]

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Activity

EXAMPLE 4 Graph a Combination of Transformations

Pose Purposeful Questions ETP

PART A

Q: According to the equation, how many transformations are carried out?

[There are two transformations: a horizontal translation two units left, indicated by the $+2$ to x; and a reflection over the x-axis, indicated by the -1 as the coefficient of $f(x)$.]

Q: How is the reflection created after the translation has occurred?

[Each point on the reflected image is exactly the same distance below the x-axis as the corresponding point of the original image is above the x-axis.]

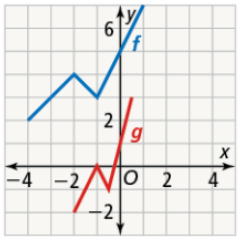
PART B

Q: For the vertical compression, what distance is affected?

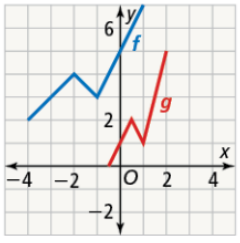
[Each point of the original function will be $\frac{1}{2}$ the distance from the x-axis for the vertically compressed image.]

Try It! Answers

4. a.



b.

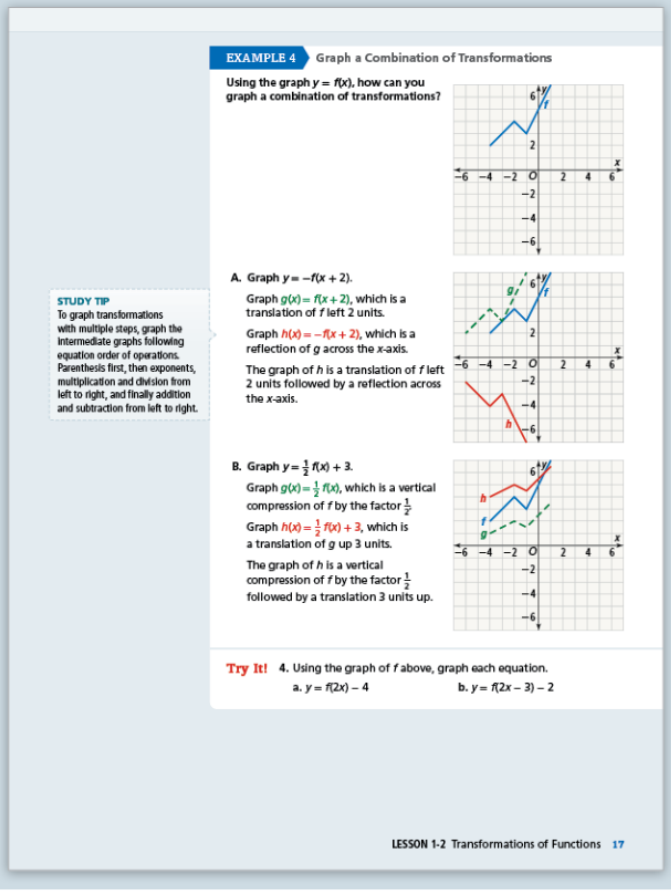


Elicit and Use Evidence of Student Thinking ETP

Q: In part (b) what order would be best for carrying out the transformations?

[$f(2x - 3) - 2 = f(2(x - 1.5)) - 2$, so apply the horizontal compression by a factor of 2 first, then the horizontal shift right 1.5 units, and last, the vertical shift down 2 units.]

Examples are available at SavvasRealize.com.



EXAMPLE 4 Graph a Combination of Transformations

Using the graph $y = f(x)$, how can you graph a combination of transformations?

STUDY TIP
To graph transformations with multiple steps, graph the intermediate graphs following equation order of operations. Parenthesis first, then exponents, multiplication and division from left to right, and finally addition and subtraction from left to right.

A. Graph $y = -f(x + 2)$.

Graph $g(x) = f(x + 2)$, which is a translation of f left 2 units.

Graph $h(x) = -f(x + 2)$, which is a reflection of g across the x-axis.

The graph of h is a translation of f left 2 units followed by a reflection across the x-axis.

B. Graph $y = \frac{1}{2}f(x) + 3$.

Graph $g(x) = \frac{1}{2}f(x)$, which is a vertical compression of f by the factor $\frac{1}{2}$

Graph $h(x) = \frac{1}{2}f(x) + 3$, which is a translation of g up 3 units.

The graph of h is a vertical compression of f by the factor $\frac{1}{2}$ followed by a translation 3 units up.

Try It! 4. Using the graph of f above, graph each equation.

a. $y = f(2x) - 4$ b. $y = f(2x - 3) - 2$

LESSON 1-2 Transformations of Functions 17

Use with EXAMPLES 3 & 4

HABITS OF MIND

Model with Mathematics If the graph of a parent function is vertically stretched by a factor of 2 and then translated 3 units down, would you get the same graph if you translated the parent graph 3 units down first and then vertically stretched it by a factor of 2? Explain.

[No; the order the transformations are performed in affects the final result.]

EXAMPLE 5 Identify Transformations From an Equation

Use and Connect Mathematical Representations ETP

PART A

Q: How are vertical transformations expressed differently in function notation than horizontal transformations?

[When $f(x)$ is written as $af(x) + k$, a signals a vertical stretch or compression and k signals a vertical shift. When $f(x)$ is written as $f[b(x - h)]$, b signals a horizontal stretch or compression and h signals a horizontal shift.]

PART B

Q: What can you tell about the shape of the graph of g compared to the shape of the graph of f, just by looking at the equation for g? Explain.

[The shape of the graphs will be the same because the equation for g shows only a horizontal and a vertical translation, with no stretch or compression.]

Try It! Answers

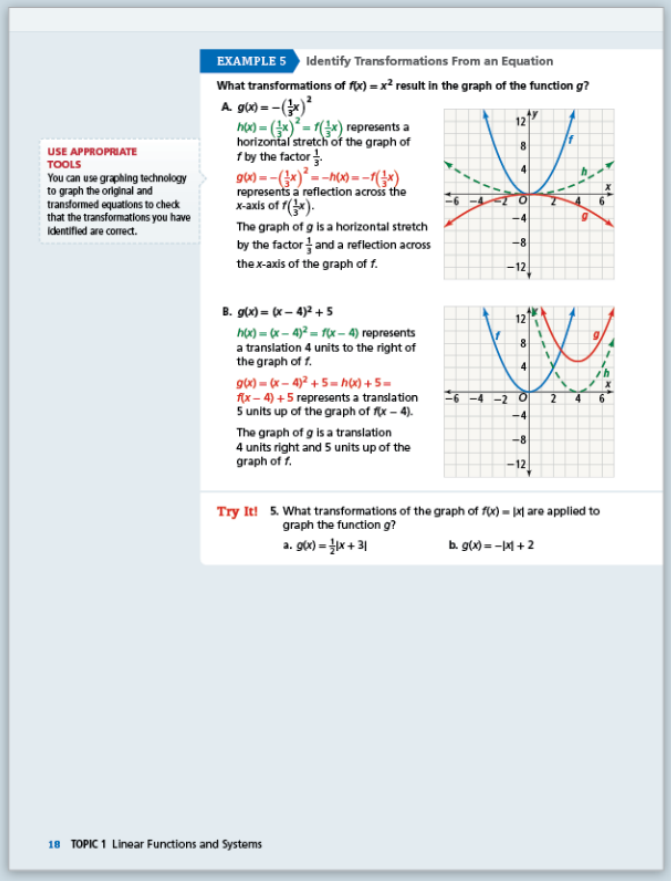
5. a. translation left 3 units and then vertical compression by the factor $\frac{1}{2}$
   b. reflection across the x-axis and then translation up 2 units

Elicit and Use Evidence of Student Thinking ETP

Q: What determines whether the function is transformed vertically or horizontally?

[Vertical transformations are indicated by operations performed on the output of the function; horizontal transformations are indicated by operations performed on the input of the function.]

Examples are available at SavvasRealize.com.



EXAMPLE 5 Identify Transformations From an Equation

What transformations of $f(x) = x^2$ result in the graph of the function g?

USE APPROPRIATE TOOLS
You can use graphing technology to graph the original and transformed equations to check that the transformations you have identified are correct.

A. $g(x) = -\left(\frac{1}{3}x\right)^2$

$h(x) = \left(\frac{1}{3}x\right)^2 = f\left(\frac{1}{3}x\right)$ represents a horizontal stretch of the graph of f by the factor $\frac{1}{3}$.

$g(x) = -\left(\frac{1}{3}x\right)^2 = -h(x) = -f\left(\frac{1}{3}x\right)$ represents a reflection across the x-axis of $f\left(\frac{1}{3}x\right)$.

The graph of g is a horizontal stretch by the factor $\frac{1}{3}$ and a reflection across the x-axis of the graph of f.

B. $g(x) = (x - 4)^2 + 5$

$h(x) = (x - 4)^2 = f(x - 4)$ represents a translation 4 units to the right of the graph of f.

$g(x) = (x - 4)^2 + 5 = h(x) + 5 = f(x - 4) + 5$ represents a translation 5 units up of the graph of $f(x - 4)$.

The graph of g is a translation 4 units right and 5 units up of the graph of f.

Try It! 5. What transformations of the graph of $f(x) = |x|$ are applied to graph the function g?

a. $g(x) = \frac{1}{2}|x + 3|$ b. $g(x) = -|x| + 2$

18 TOPIC 1 Linear Functions and Systems

SavvasRealize.com
Activity

EXAMPLE 6 Write an Equation From a Graph

Pose Purposeful Questions ETP

Q: What do you notice about the graph of this function when compared to the parent function?
[The graph of the parent function has been reflected vertically, shifted to the right, and shifted up.]

Q: How can you verify the value of *a* from the graph?
[If the value of *a* is 1, the graph of the parent function can be placed on top of the transformed function to match its shape.]

Try It! Answer

6. There would be a vertical stretch by a factor of 2. The new equation would be $y = 2(-|x - 15| + 15) = -2|x - 15| + 30$.

Elicit and Use Evidence of Student Thinking ETP

Q: How can the phrase "traveled twice as far in the same amount of time" be interpreted with regard to the domain and range?
[The domain will stay the same because there was no change in time, and the range will be doubled.]

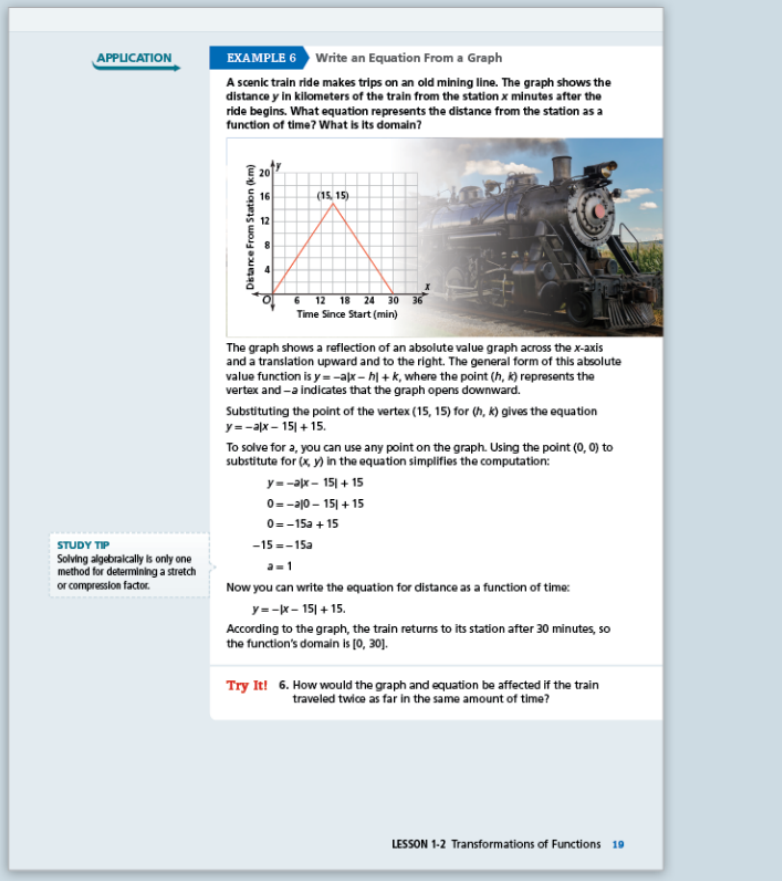
Use with EXAMPLES 5 & 6

HABITS OF MIND

Make Sense and Persevere The function $f(x) = |x|$ is translated 3 units right and 2 units down, and then vertically stretched by a factor of 4. What is an equation for the transformed function g?

[$g(x) = 4(|x - 3| - 2)$, or $g(x) = 4|x - 3| - 8$]

Examples are available at SavvasRealize.com.



APPLICATION

EXAMPLE 6 Write an Equation From a Graph

A scenic train ride makes trips on an old mining line. The graph shows the distance *y* in kilometers of the train from the station *x* minutes after the ride begins. What equation represents the distance from the station as a function of time? What is its domain?

The graph shows a reflection of an absolute value graph across the *x*-axis and a translation upward and to the right. The general form of this absolute value function is $y = -a|x - h| + k$, where the point (h, k) represents the vertex and $-a$ indicates that the graph opens downward.

Substituting the point of the vertex (15, 15) for (h, k) gives the equation $y = -a|x - 15| + 15$.

To solve for *a*, you can use any point on the graph. Using the point (0, 0) to substitute for (x, y) in the equation simplifies the computation:

$y = -a|x - 15| + 15$
$0 = -a|0 - 15| + 15$
$0 = -15a + 15$
$-15 = -15a$
$a = 1$

STUDY TIP
Solving algebraically is only one method for determining a stretch or compression factor.

Now you can write the equation for distance as a function of time:

$y = -|x - 15| + 15$.

According to the graph, the train returns to its station after 30 minutes, so the function's domain is [0, 30].

Try It! 6. How would the graph and equation be affected if the train traveled twice as far in the same amount of time?

LESSON 1-2 Transformations of Functions 19

Extend Student Thinking

USE WITH EXAMPLE 6 Students use what they know about writing an equation from a graph to practice writing an equation from context.

- Another train ride is available that lasts 45 minutes and travels a total distance of 25 km.

Q: Explain how to write an equation to model this situation.
[Because the total time is 45 min and the total distance is 25 km, the turn-around point, or vertex, will occur at (22.5, 12.5). Plug the values of the vertex into the equation for *h* and *k*, along with the point (0, 0), and solve for *a*.]

Q: What equation represents the distance from the station as a function of time?
[$d = -\frac{5}{9}|m - 22.5| + 12.5$]

SavvasRealize.com
Presentation Practice

CONCEPT SUMMARY Transformations of Functions

Q: What happens to the graph of $f(x) = a \cdot f[b(x - h)] + k$ as you change the values of a, b, h, and k?

[Changing a results in a vertical stretch or compression. Changing b results in a horizontal stretch or compression. Changing h results in a horizontal translation. Changing k results in a vertical translation.]

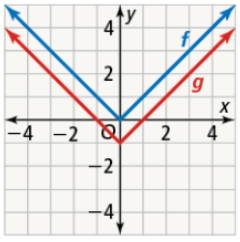
Do You UNDERSTAND? | Do You KNOW HOW?

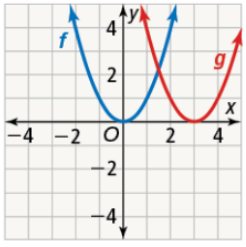
Common Error

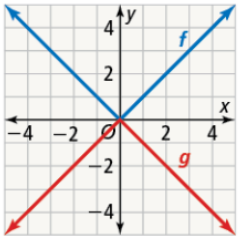
Exercise 5 Students may incorrectly interpret the negative sign in the function g as a translation of 3 units to the left. When students are unsure, suggest that they make a table of values to check the points on their graph.

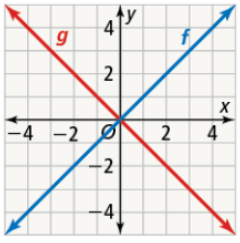
Answers

1. From the equations, you can see whether the graph of the parent function will be shifted up or down, reflected over an axis, or stretched or compressed to create the graph of the new function.
2. $g(x) = f(x) + k$ is a vertical translation k units, so the output of the function is affected. $g(x) = f(x - h)$ is a horizontal translation h units, so the input of the function is affected.
3. $g(x)$ is a horizontal translation of $f(x)$ left 1 unit, not right 1 unit, because $h < 0$.

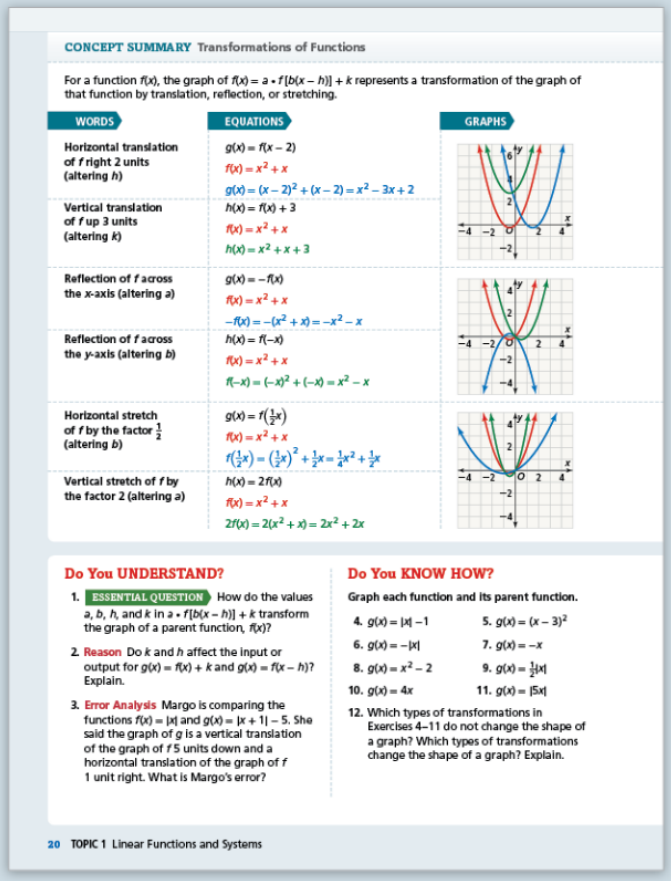
4.


5.


6.


7.


Concept Summary, Do You Understand, and **Do You Know How** are available at SavvasRealize.com.


CONCEPT SUMMARY Transformations of Functions

For a function $f(x)$, the graph of $f(x) = a \cdot f[b(x - h)] + k$ represents a transformation of the graph of that function by translation, reflection, or stretching.

WORDS	EQUATIONS	GRAPHS
Horizontal translation of f right 2 units (altering h)	$g(x) = f(x - 2)$ $f(x) = x^2 + x$ $g(x) = (x - 2)^2 + (x - 2) = x^2 - 3x + 2$	
Vertical translation of f up 3 units (altering k)	$h(x) = f(x) + 3$ $f(x) = x^2 + x$ $h(x) = x^2 + x + 3$	
Reflection of f across the x-axis (altering a)	$g(x) = -f(x)$ $f(x) = x^2 + x$ $-f(x) = -(x^2 + x) = -x^2 - x$	
Reflection of f across the y-axis (altering b)	$h(x) = f(-x)$ $f(x) = x^2 + x$ $f(-x) = (-x)^2 + (-x) = x^2 - x$	
Horizontal stretch of f by the factor $\frac{1}{2}$ (altering b)	$g(x) = f\left(\frac{1}{2}x\right)$ $f(x) = x^2 + x$ $f\left(\frac{1}{2}x\right) = \left(\frac{1}{2}x\right)^2 + \frac{1}{2}x = \frac{1}{4}x^2 + \frac{1}{2}x$	
Vertical stretch of f by the factor 2 (altering a)	$h(x) = 2f(x)$ $f(x) = x^2 + x$ $2f(x) = 2(x^2 + x) = 2x^2 + 2x$	

Do You UNDERSTAND?

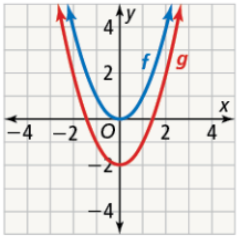
1. ESSENTIAL QUESTION How do the values a, b, h, and k in $a \cdot f[b(x - h)] + k$ transform the graph of a parent function, $f(x)$?
2. Reason Do k and h affect the input or output for $g(x) = f(x) + k$ and $g(x) = f(x - h)$? Explain.
3. Error Analysis Margo is comparing the functions $f(x) = |x|$ and $g(x) = |x + 1| - 5$. She said the graph of g is a vertical translation of the graph of f 5 units down and a horizontal translation of the graph of f 1 unit right. What is Margo's error?

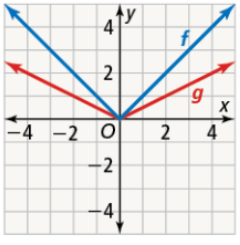
Do You KNOW HOW?

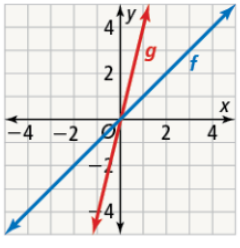
Graph each function and its parent function.

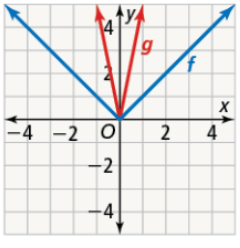
4. $g(x) = |x| - 1$
5. $g(x) = (x - 3)^2$
6. $g(x) = -|x|$
7. $g(x) = -x$
8. $g(x) = x^2 - 2$
9. $g(x) = \frac{1}{2}|x|$
10. $g(x) = 4x$
11. $g(x) = |5x|$
12. Which types of transformations in Exercises 4–11 do not change the shape of a graph? Which types of transformations change the shape of a graph? Explain.

20 TOPIC 1 Linear Functions and Systems

8.


9.


10.


11.


12. Vertical and horizontal translations and reflections do not change the shape of a graph. Stretches and compressions change the shape of a graph because they affect the steepness of the graph.

PRACTICE & PROBLEM SOLVING

Practice & Problem Solving and **video tutorials** are available at SavvasRealize.com.

Lesson Practice You may opt to have students complete the automatically scored Practice and Problem Solving items online powered by MathXL for School.

Choose from: **Lesson Practice**

Adaptive Practice

You may also take advantage of the bank of exercises for assigning additional practice.

Assignment Guide

On-Level	Advanced
14, 16–38	13–38

Engage Through Student Choice

Promote student agency by allowing students to choose practice items. You may structure this choice in many ways.

For example:

Assign each section a point value. Students choose at least one item from each section and items chosen should have a minimum of 20 total points.

Understand, Apply.....2 points each

Practice.....1 point each

Assessment Practice.....1 point each

Performance Task.....3 points

Item Analysis

Example	Items	DOK
1	18–21	1
	13	3
2	16, 22, 23	1
	14	2
3	24–27	1
	15	3
4	28	1
5	29–32, 36	2
	17, 37	3
6	33–35, 38	3

Scan the QR code to access a video tutorial.

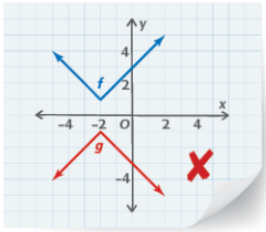
PRACTICE & PROBLEM SOLVING

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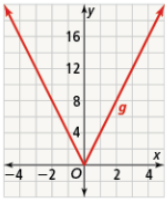
UNDERSTAND

13. Use Structure Write a function g with the parent function $f(x) = x^2$ that has a vertex at (3, –6). $g(x) = (x - 3)^2 - 6$

14. Error Analysis Describe and correct the error a student made in graphing $g(x) = f(-x)$ as a reflection across the y-axis of the graph of $f(x) = |x + 2| + 1$.



15. Higher Order Thinking Describe the transformation g of $f(x) = |x|$ as a stretch and as a compression. Then write two equations to represent the function. What can you conclude? Explain.



16. Use Structure The graph of the parent function $f(x) = x^2$ is reflected across the y-axis. Write an equation for the function g after the reflection. Show your work. Based on your equation, what happens to the graph? Explain.

17. Error Analysis Monisha is comparing $f(x) = |x|$ and $g(x) = |2x - 4|$. She said the graph of g is a horizontal translation of the graph of f 4 units to the right and a horizontal compression of the graph of f by a factor of 2. What is Monisha's error?

PRACTICE

Graph each function as a translation of its parent function, f. How did the transformation affect the domain and range? SEE EXAMPLE 1

18. $g(x) = |x| - 5$ **19.** $g(x) = (x + 1)^2$

20. $g(x) = |x - 3|$ **21.** $g(x) = x^2 + 2$

What is the function for the image graph? Check by graphing. SEE EXAMPLE 2

22. Reflect $f(x) = x^2 + 1$ across the x-axis. $f(x) = -x^2 - 1$

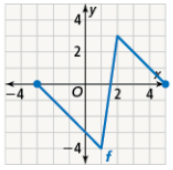
23. Reflect $f(x) = x^2 + 1$ across the y-axis. $f(x) = x^2 + 1$

Graph each function as a vertical stretch or compression of its parent function. SEE EXAMPLE 3

24. $g(x) = 0.25|x|$ **25.** $g(x) = 3x^2$

26. $g(x) = 1.5|x|$ **27.** $g(x) = 0.75x^2$

28. Use the graph of $f(x)$ to graph $y = f(x + 1) + 2$. SEE EXAMPLE 4



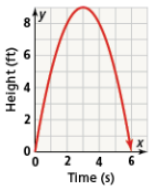
What transformations of $f(x) = x^2$ are applied to the function g? Check by graphing. SEE EXAMPLE 5

29. $g(x) = 2(x + 1)^2$ **30.** $g(x) = (x - 3)^2 + 5$

31. $g(x) = -x^2 - 6$ **32.** $g(x) = 4(x - 7)^2 - 9$

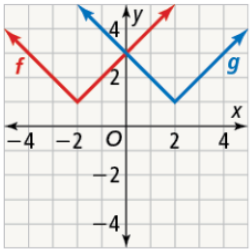
33. The graph shows the height y in feet of a flying insect x seconds after taking off from the ground. Write an equation that represents the height of the insect as a function of time. SEE EXAMPLE 6

$y = -(x - 3)^2 + 9$



Answers

14. The student reflected the graph of $f(x)$ across the x-axis instead of reflecting the graph across the y-axis.



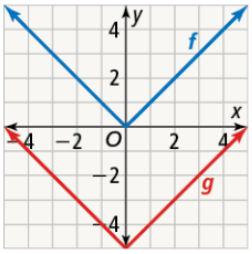
24–32, 34. See back of book.

15. Multiplying the y-values of the points on the graph of f by 4 results in points on the graph of g. So, $g(x)$ is a vertical stretch of $f(x)$ by a scale factor of 4 and can be written as $g(x) = 4|x|$. Dividing the x-values of the points on the graph of f also results in points on the graph of g. So $g(x)$ is a horizontal compression of $f(x)$ by a scale factor of 4 and can be written as $g(x) = |4x|$. A vertical stretch is the same as a horizontal compression in this case.

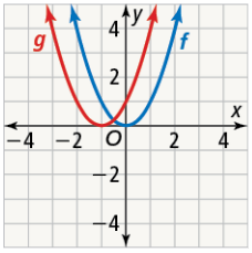
See answers for Exercises 16–21 on the next page.

16. $g(x) = f(-x) = (-x)^2 = (x)^2$; The graph of $g(x)$ is the same as the graph of $f(x)$.

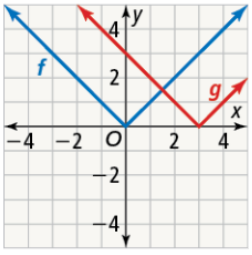
17. $g(x) = |2x - 4|$ is equivalent to $g(x) = 2|x - 2|$. So, $g(x)$ is a horizontal translation of $f(x)$ right 2 units and a vertical stretch of $f(x)$ by a factor of 2.

18. 

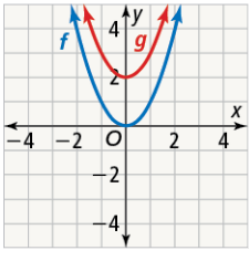
The domain values of f and g are the same, but the range values of g are 5 units less than the range values of f.

19. 

The range values of f and g are the same, but the domain values of g are 1 unit less than the domain values of f at corresponding range values.

20. 

The range values of f and g are the same, but the domain values of g are 3 units greater than the domain values of f at corresponding range values.

21. 

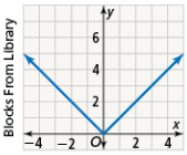
The domain values of f and g are the same, but the range values of g are 2 units greater than those of f.

Practice & Problem Solving is available at SavvasRealize.com.

PRACTICE & PROBLEM SOLVING

APPLY

34. **Model With Mathematics** Chiang walks to school each day. She passes the library halfway on her walk to school. She walks at a rate of 1 block per minute. The graph shows the distance Chiang is from the library as she walks to school.

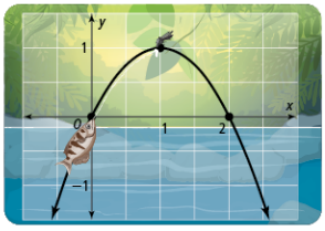


a. Write a function, f, to model the distance Chiang is from the library when she walks to school.

b. If Chiang jogs to school, she travels at a rate of 2.5 blocks per minute. Write a function, g, to model the distance Chiang is from the library when she jogs to school.

c. Graph the function, g, that models the distance Chiang is from the library when she jogs to school.

35. **Model With Mathematics** The archer fish spits stream of water at flying insects to knock them into the lake. The path of the stream of water is shown with x and y distances in feet. Write an equation to represent the path of the stream of water in relation to the coordinate grid. Then determine the coordinates of the point of maximum height of the stream of water.



ASSESSMENT PRACTICE

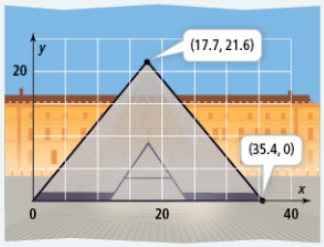
36. Match each equation on the left to an equation on the right that has the same translation of $y = |x|$.

A. $y = |x| - 1$ — D. $y = |x + 1| - 1$
B. $y = -|x + 1|$ — E. $y = 2|x| - 1$
C. $y = -|x + 1| - 1$ — F. $y = |x + 1|$

37. **SAT/ACT** Which translation is part of transforming $f(x) = x^2$ into $h(x) = (x + 4)^2 - 2$?

● left 4 units
Ⓒ right 2 units
Ⓑ left 2 units
Ⓓ right 4 units

38. **Performance Task** The Louvre Pyramid in Paris is shown on the coordinate grid, where x and y are measured in meters and the ground is represented by the x-axis.



Part A The outline of the Pyramid is a transformation of the function $f(x) = |x|$. Write a function g to model the outline of the Pyramid.

Part B What is the domain and range of the function that models the outline of the Pyramid? What do the domain and range represent?

35. $y = -(x - 1)^2 + 1$; (1, 1)

38. **Part A** $g(x) = -1.22|x - 17.7| + 21.6$

Part B domain: $0 \le x \le 35.4$; range: $0 \le y \le 21.6$; The domain represents the width of the pyramid, which is 35.4 m wide. The range represents the height of the pyramid, which is 21.6 m high.

24–32, 34. See back of book.

LESSON QUIZ

Use the Lesson Quiz to assess students' understanding of the mathematics in the lesson.

Students can take the Lesson Quiz online or you can download a printable copy from **SavvasRealize.com.** The Lesson Quiz is also available in the *Assessment Sourcebook*.

Item Analysis

Item	DOK	Skills Review and Practice
1	2	F68
2	1	F57
3	1	F57
4	2	F56
5	2	F25, F57

RtI Use the student scores on the Lesson Quiz to prescribe differentiated assignments.

If students take the Lesson Quiz online, it will be automatically scored and appropriate differentiated practice will be assigned based on student performance.

I Intervention	0–3 points	• Reteach to Build Understanding • Mathematical Literacy and Vocabulary • Lesson Virtual Nerd videos
O On-Level	4 points	• Enrichment
A Advanced	5 points	• Enrichment

Lesson Quiz is available at SavvasRealize.com.

ASSESSMENT RESOURCES

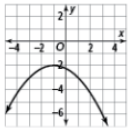
Name _____

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1-2 Lesson Quiz

Transformations of Functions

1. Select all the transformations of $f(x) = x^2$ that combine to result in the graph of function g at the right.



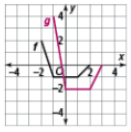
- [x] A. translation of 1 unit left
- [ ] B. translation of 1 unit right
- [x] C. translation of 2 units down
- [x] D. horizontal stretch by a factor of 2
- [ ] E. vertical stretch by a factor of 2
- [x] F. reflection across the x-axis

2. What is the equation for g, which is $f(x) = 2x^2 + 3x - 1$ reflected across the y-axis?

Ⓐ $g(x) = 2x^2 + 3x - 1$ Ⓒ $g(x) = 2x^2 - 3x - 1$

Ⓑ $g(x) = -2x^2 - 3x + 1$ Ⓓ $g(x) = -2x^2 - 3x - 1$

3. The graph represents f. Graph $g(x) = 2f(x - 1)$.

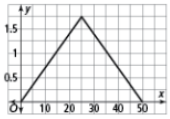


4. A bucket has a leak that started 6 hours ago. At time 0, when the bucket just emptied, a tap was turned on for 3 hours. The amount of water in the bucket can be modeled by the function $f(x) = |x|$ over the interval $[-6, 3]$, where x is the time, in hours, and y is the amount of water in the bucket, in gallons. A similarly full bucket started to leak only 3 hours ago. Which translation of the graph represents the amount of water in the second bucket if it is moved under the tap as soon as it empties?

Ⓐ $g(x) = |x| + 3$ Ⓒ $g(x) = |x| - 3$

Ⓑ $g(x) = |x + 3|$ Ⓓ $g(x) = |x - 3|$

5. The graph of the function f is shown. The domain of f is $[0, 50]$. What is the range of $g(x) = 4f(x)$?



Ⓐ $[0, 50]$ Ⓒ $[0, 1.75]$

Ⓑ $[0, 70]$ Ⓓ $[0, 7]$

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